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AMERICAN HAND SIGN LANGUAGE CLASSIFICATION

A Project Report

Submitted in partial fulfilment of the
requirements for the award of the Degree of

BACHELOR OF SCIENCE (DATA SCIENCE)

By

Kapish Shukla

Kunal Solanki

Seat Number: _____

Under the esteemed guidance of

Ms. Kanchan Taksale

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DEPARTMENT OF INFORMATION TECHNOLOGY AND DATA SCIENCE

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

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MUMBAI, 400 037

MAHARASHTRA

2024 - 2025

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This is to certify that the project entitled, "**American Hand Sign Language Classification**",
is bonafied work of **KAPISH SHUKLA AND KUNAL SOLANKI** bearing Seat No:
_____ submitted in partial fulfilment of the requirements for the award of
degree of BACHELOR OF SCIENCE in DATA SCIENCE from University of Mumbai.

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ABSTRACT

5 The challenges faced by individuals who are both deaf and mute when attempting to
5 communicate with others stem from the disparate levels of knowledge about American Sign
Language (ASL) among the general population. This results in a boundary to communication
and therefore a barrier to expressing feelings or needs during everyday living. Our objective
was to create a high-tech system capable of interpreting hand signs and communicating their
meaning to everyone. Thus, technology becomes the bridge to help unite different individuals.

We created the ability to recognize hand signs by using thousands of photographs of
letters A, B, C, D, and E, taken from many different perspectives and locations. Python and
machine learning-based tools (such as TensorFlow) were used to create a "brain" for the
computer called a Convolutional Neural Network (CNN). Before we began training, we
standardized the size of all images, as well as cleaned up all imagery, to provide the best
scenario possible for the computer to learn. This has absolutely affected how quickly the
computer will be able to recognize hands.

From our findings, we were able to judge the accuracy of our CNN model and determine
that it is performing quite well and producing accurate results when tested with a variety of
hand gestures. It is amazing that a computer is now capable of interpreting a person's
communication through gestures. This proves that our idea works and can be very useful for
society.

3 In the future, we want to add all the letters of the alphabet and even full sentences to
our system. We also plan to make a mobile app so people can use it anywhere on their phones.
Our main goal is to make a real-time system that works with live video and cameras. We hope
this project helps many people live a better and easier life.

ACKNOWLEDGEMENT

I take this opportunity to thank all those who helped me during my semester project. Otherwise, it would have been impossible without their great help and advice.

I should thank Vidyalankar School of Information Technology for offering me this opportunity to do the work and for establishing an amiable learning atmosphere that promotes newness and research. I will particularly enjoy the resources and support I will receive from the institution during the period of my project.

First and foremost, I would like to put on record the special gratitude I have in my heart for my project guide, Prof. Shajil Kumar; without all his advice, encouragement, and feedback during the project, much of this couldn't have been possible. He has been an inspiring mentor and an endless source of inspiration. All the expertise and knowledge in the fields of e-commerce and machine learning have been crucial for me to face challenges and enhance the quality of work.

I would like to extend my gratitude to all faculty members of Vidyalankar School of Information Technology. Their teaching and guidance have contributed significantly to my learning. Special thanks to the administrative staff for their assistance and to fellow students for valuable discussion and support throughout this journey.

DECLARATION

I hereby declare that the project entitled, “**AMERICAN HAND SIGN LANGUAGE CLASSIFICATION**” done at Vidyalankar School of Information Technology, has not been in any case duplicated to submit to any other universities for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (DATA SCIENCE)** to be submitted as final semester project as part of our curriculum.

Name and Signature of the Student

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Chapter 1 Introduction

Communication is very important for everyone to talk and share ideas. But for people who cannot speak or hear, it is very difficult because they use American Sign Language (ASL). Most normal people don't understand these hand signs, so there is a big gap in communication.

We want to make a solution to the problem with technology so everyone will understand it easily. For our project, we use Computer Vision and Deep Learning to track hand gestures (like how to do sign language). Our ultimate goal is to create a system that will look at a picture of someone's hand and tell them what letter it is making. This will help create a bridge between the hearing world of those who cannot communicate with hearing-impaired people, so everyone can communicate with each other ultimately making the whole world a better place to communicate.

To do this, we are using Convolutional Neural Networks (CNN) to help us teach the computer how to visually identify different hands doing different letters. These types of systems work similarly to how our brain does when we see an object or look at something; they are able to learn over time. We used lots of hand gesture images off the internet to train our model. Once we have done that, our computer will have a very good ability to recognize sign language, even though different lighting conditions or backgrounds can be present during the video; therefore, the model will have a very good recognition of sign language.

These days, artificial intelligence is being used in various areas to support individuals in no matter the background they are from; whether it's in health care or education, our project (like many) is helping society using these newer technologies. We hope to one day have a genuine application of our project on each mobile phone. We feel that it's very important to help individuals through projects like ours because artificial intelligence will ultimately support the world!

1.1 Background

1.1.1 Overview

This project focuses on assisting individuals who are unable to communicate verbally through the use of hand gestures/signing. We have developed a smart computer system that is able to take images of a person using American Sign Language and identify the correct letter associated with the gesture. Currently, our model can successfully classify A, B, C, D, and E at a high accuracy level. We acquired thousands of images from Kaggle to teach the computer how hand signs look like in various lighting environments. Essentially, we gave a computer vision so that it can learn how to interpret the silent mode of communication for deaf individuals.

For the development of this system, we utilized Python and a technology referred to as CNNs (Convolution Neural Networks). We collected thousands of hand sign images, cleaned them up (resized to proper dimensions, made sure that they were visually clear) enough that the computer could analyze them without issue, and trained our model to correctly interpret the patterns associated with the hand sign. Ultimately, we would like the computer to not make any errors when encountering a new sign. In the future, we want to make this into a real app that can help everyone communicate without any problems.

1.1.2 Rationale for choosing this topic

I chose this topic because I saw that many people who cannot speak or hear face big problems in their daily life. They use hand signs to talk, but most of us don't understand what they are saying. I felt very sad about this communication gap and wanted to do something to help. By using technology, we can make a bridge so that everyone can understand their feelings and needs easily.

Another reason is my interest in how computers can see and understand things like humans. I wanted to learn how Artificial Intelligence and Python can solve real-world problems. This project is a perfect way to use my coding skills for a good cause. It makes me happy to think that my small project can one day help a person communicate with the whole world.

1.1.3 Significance

The significance of this project is very big because it helps people who cannot talk normally. Many times, deaf people feel alone because nobody understands their hand signs in shops or hospitals. Our system can quickly translate these signs into text so everyone can

understand them easily. This will give them more confidence and make them feel like a part of our society.

Another important thing is that this project uses new technology like AI to do something good for humans. Computers are not solely used for gaming — they can also aid human beings by solving relevant challenges. Eventually, it will be possible for schools to employ this technology to teach other students how to utilize sign language effectively. Furthermore, it offers fast and easy means of communication to everyone without requiring human translators on a constant basis.

1.1.4 Challenges

There were multiple challenges associated with the development of this project. The primary issue has been the variable conditions of exposure and placement of subject images (background) within photographs. For example, the computer module cannot identify hand signs when the image was taken in a dimly lit environment or if other objects were located behind the signer's hand. In order to overcome this issue, significant amounts of time were required to clean the images to ensure the object for processing the image could focus solely on the hand(s) of the subject (signer).

Another challenge stemmed from the fact that some hand signs are so similar in their physical appearance that one cannot distinguish between them. For example, the hand sign for "A" and the hand sign for "E" look very much like one another based on an imperfect camera angle. Consequently, training the CNN model to identify these subtle distinctions was very difficult. Additionally, collecting and processing the thousands of images necessary to train the CNN model required a significant amount of time and computing power.

1.1.5 Justification

This project will allow deaf people to communicate in real time with their friends and family who do not know sign language; however, it is the first technology of its kind to enable this communication. Our goal with this project is to demonstrate how simple cameras combined with artificial intelligence (AI) can successfully detect sign language in an easy-to-implement and cost-effective manner that can significantly improve the quality of life for individuals who cannot speak.

In addition, we will utilize the Convolutional Neural Network (CNN) model, which is one of the most accurate models available for image recognition. By using an established image recognition algorithm, we establish that our solution can move beyond demonstration of capability into a truly working solution that can be continuously improved upon. Additionally,

leveraging Python and publicly available data allows for easy replication of our technology by other developers who can further develop this idea through their own efforts. We encourage other developers to use projects such as this one to apply their knowledge of science to benefit society and create more unity within the community.

1.1.6 Historical Methods

Overview of Historical Sign Recognition -

Historically it was impossible to have a conversation with someone who was deaf because they did not have the benefit of technology that would allow them to communicate with hearing individuals efficiently. Although some individuals attempted to either write down examples of sign language, or utilize instructional books to teach others how to use sign language, it was extremely time consuming and too tedious for most people to continue to use these methods of communication. As a result, deaf individuals and hearing individuals could not communicate with one another for a very long time; no technology existed to help computers be able to recognize or interpret the signs that deaf individuals used.

Eventually, scientists began to develop very rudimentary cameras combined with basic computer algorithms in an attempt to have computers recognize the shapes of hand signs used by deaf individuals, however these attempts were unsuccessful when using a camera to recognize shapes in a dark room or a room that had too much clutter. Fortunately, advancements in technology, specifically using **Deep Learning and Convolutional Neural Networks**, have allowed computers to recognize hand signs similarly to the way an individual with normal visual acuity would. This historical account illustrates how communication methods have transitioned over time from printed books containing examples of sign language to artificial intelligence based systems designed to assist with disabilities.

Challenges of Historical Methods

The previous methods for recognizing sign language contained severe issues. Computers were not able to recognize physical hand forms through an image. Special electronic gloves were required to help track hand motions (the gloves contain many wires). These gloves were also difficult to wear, as they were costly and created additional strain due to their weight. Therefore, it was extremely challenging for the average person to use such elaborate devices to say "hello" or "thank you."

Other difficulties involve the previous design of the computer systems. The systems only work in plain backgrounds and in perfect lighting conditions. Thus, if a person moved or

the room was somewhat dark, then the system would fail entirely. Many of the previous systems could recognize only a small number of signs and were also very slow in providing results. This is why previous technologies were not practical for real-world application and why we need AI today.

1.1.7 Modern Advancements Innovations in Technology

Recent changes in our lives due to improving technology include new developments in artificial intelligence and deep learning. In the past, computers could only understand simple instructions, but now with CNN, computers can "see" and "think" like humans. This means that our project can use a simple camera to recognise hand gestures and do so without the need for any additional wires or expensive equipment. Everything we had to go through before to get to this point has made it so much easier and faster than before.

Another big innovation that has come along recently is the ability to store and process thousands of images in just minutes with cloud computing. This was not possible for an average student to train a model with so many images before, but it now is very environmentally friendly to create large amounts of data on their own systems and through the internet. These new technologies also help us to create applications that can assist persons with disabilities to communicate with each other in real time. With these smart innovations, we can envision future communication without any barriers of language.

Advantages of Modern Techniques

State-of-the-art methods such as Convolutional Neural Networks (CNNs) and Deep Learning provide very high levels of accuracy and speed in recognizing hand signs being made. Due to the advances in technology, there is no longer a need for special gloves or expensive camera sensors to recognize hand signs. A standard webcam or mobile camera can now be used to detect hand gestures, thereby making the system very user-friendly to users and effective even with minimal amounts of movement in making the sign.

A further benefit of using state-of-the-art models is that they are able to learn from thousands of samples, so once the model has been trained to recognize a certain hand gesture, it will be able to correctly identify that hand gesture in different types of environments (whether illuminated or shadowed), and easily adapt to different sizes and shapes of hands when the sign

is made. Therefore, our project represents a very strong and reliable means of assisting the deaf community in their everyday lives, without the potential for error.

1.2 Objectives

Objective 1: Recognize and Classify Hand Signs

Our goal with this project is to develop a computer to recognize ASL hand signs. We want to create a smart system that can take an image of a hand and determine quickly if it's representing the letter A, B, C, D, or E. By applying machine learning, we will automate this process so that no one needs to interpret everything by hand.

Objective 2: Help People Communicate Better

Another primary goal is to allow deaf/mute individuals to have conversations with others without fear. So our project will act as an intermediary or bridge between sign language and spoken language, enabling all people to communicate using signs and hear what is being said. The system will produce a text description of each sign while making it simple for both students and the general public to learn and apply sign language in their everyday life.

Objective 3: Use Smart Technology for Good

We are utilizing modern technologies such as Python and CNN to find a solution to an actual problem affecting millions of humans per year, and we strive to enhance their quality of life. This goal involves training a "computer brain" to detect patterns in hand shapes, as humans can. The long-term goal is to provide users with an efficient, consistent, and reliable system that functions correctly and seamlessly in many locations and environments, ultimately providing deaf/mute individuals the opportunity to lead full lives like everyone else.

1.3 Purpose, Scope, Applicability (Feasibility Study)

1.3.1 Purpose

The main aim of this project is to create a smart device for the deaf and mute populations, which facilitates fast and efficient communication between all people. The design of the device is going to be a very simple camera plus a computer that can interpret hand sign language and convert them into text for others to read. With this tool, someone who does not know sign language will still be able to understand what another person is communicating.

1.3.2 Scope

We are going to use modern technologies including AI in order to eliminate the gap of communication and create an inclusive that Everyone feels like they belong; therefore, the scope of the project includes: the design and development of a machine learning-based classifier for American Sign Language hand signs from images taken by a user;

- Where the classifier will be able to extract all the significant features of the user's hand signs such as finger positioning, shape of the hands and the pattern of the movement/pattern of fingers
- Utilize smart CNN methods to analyze these visual features automatically.
- Classify the hand signs into distinct categories based on their shapes (like letters A, B, C, D, and E).
- Support up to five different alphabetical signs associated with the American Sign Language tradition.
- Deliver information about gesture recognition by computers which can support research and create better communication tools for the Deaf and Mute Population.

At this time, the scope of this program will be limited to being able only to analyze static pictures of the hand signs, therefore the entire body of sign language videos, and more than just one sign language sign/video, are beyond the scope of this project.

1.3.3 Applicability

This can be useful in several areas but particularly the following:

- Schooling in Special Education: Teachers and students who teach or learn in special schools can use this technology to make it easier for them to learn and use American Sign Language (ASL).

- **Healthcare and Hospitals:** Doctors and nurses can communicate better with patients who are deaf or mute by using this technology, which will help them provide better care.
- **Social Inclusion:** This technology will allow public places such as banks or parks to be more accessible for people with disabilities and provide an easier method of communication.
- **Assistive Technology:** This project demonstrates how we can use artificial intelligence (AI) and cameras to assist people with disabilities, and we can further develop it into a mobile app for everyday usage.

1.3.4 Significance of the System

3 The value of our system is that it will significantly improve the quality of life for the people who do not have the ability to hear or speak. Most of the time, many of those who use sign language for communication feel excluded from society since most people do not know how to sign, whereas our project will give everyone the ability to see what those using sign language are saying via a digital display on a screen. Essentially, it is a 24/7 translating device (free of charge) that will assist those who use sign language in shops, hospitals and schools. This form of technology is extremely valuable as it utilizes advanced forms of smart AI to bring all individuals together while helping to build the self-esteem of those who are deaf.

1.3.5 Technical feasibility

The success of our project depends on the availability of all of the right tools to do it technically (Python, TensorFlow) & being able to utilize as many computers & laptops as possible with sufficient power to run the CNN models that we are using to identify the hand signs. In addition, we do not need any expensive hardware -- just a basic webcam that has enough resolution to capture the images. Because there are so many datasets available on the web for sign language, training our model has also been much easier than it used to be before these datasets were created. All of this indicates that our project will be easy to build and implement in real world applications and that there should not be any significant technical problems with it.

Chapter 2 Dataset Description

2.1 Introduction

The dataset for our project is critical because it serves as the computer's "textbook". To feed our model thousands of images of various hand gestures so he/ben can learn how to identify hand signs, we have provided thousands of images of several letter(s) from the alphabet of American Sign Language; (i.e., "A", "B", "C", "D", and "E".) Without these pictures, the computer would not know what a hand looks like or how the fingers move in order to form a hand gesture.

We've made an effort to take as many different types of images in our database as possible. This means that we took images of many different people's hands with many different colors and sizes, also we used many different backgrounds and illuminating sources so the computer could become as intelligent as possible. Our theory is that if the database contains enough of a variety of images, then the model will perform better in the actual world when someone is using our model in a room or outside.

Before we can give the computer these images, we need to organize them into separate folders for the computer to know which sign corresponds to what folder - this is referred to as data collection and was our first project step. By creating a clean and sufficient image database, we can be confident that our Sign Language system will be very accurate and will produce correct results.

2.2 Data Source

Our project relied on knowledge from a popular website, Kaggle, which contains a wealth of data sets and has been created by some really brilliant individuals who have made their data available for free. The data set we downloaded contain thousands of images of hands demonstrating different letters in a specific language (American Sign Language). This data was extremely useful in making it simple for us because we would not have had to construct thousands of hand images by taking pictures of them with our camera.

In addition to using the data from Kaggle, we have also used photographs of our own hands taken with a laptop webcam to validate the accuracy of the model created to recognize different letter signs. The combination of data from both sources allowed for a much more

realistic and robust result. It was also very easy for us to download and incorporate the downloaded files into our Python application in order to begin training the computer model.

2.3 Dataset Characteristics

Our Hand Sign Language collection contains:

1. **A lot of pictures:** We have many different images of a person demonstrating each hand sign, such as A, B, C, D & E. We want our model to see different people performing all of these signs so that it can learn correctly.

2. **Features:** Each picture has different measurements, such as where the fingers are positioned and how big the palm appears. These two examples of "features" are important in defining consistency between sign letters and separating them.

The quality of our pictures is important because they come from various rooms and lighting sources. The types of images may also vary; for example, some may be zoomed in while others are not. This will create a strong model. Although our images were initially generated as simple photographs, we converted them to numbers (pixels) to allow Python to read the images easily. This variety will help us create a highly accurate and universally applicable model.

2.4 Data Features

There are multiple key features in the dataset that aid computers in identifying hand signs. We mainly examine the following attributes of each image:

- Hand Shape:** This indicates the overall form of the hand such as whether it is in a fist or an open palm.

- Finger Positions:** This is vital in determining where each finger is oriented (upward or downward) so the system can ascertain which letter is being displayed.

- Pixel Values:** The pixel value describes how bright or dark one specific point is in relation to the rest of the image so that computers can recognize hands as being separate from the background.

- Key Points:** Key points represent specific reference locations on the knuckles and tips of the fingers that are used to compute the distance between each finger.

The above listed characteristics are obtained from images using specialized code that processes the entire hand by examining each individual feature associated with it, and by

acquiring as many points as possible from each of those features being analyzed, even when the pictures might be slightly blurred, the system can still interpret what gesture corresponds directly with those variables being evaluated. Based upon how we examine these four characteristics, it is easy for the system to differentiate between the letter "A" and the letter "B".

2.5 Dataset Usage

To work with the data, we split all images in the dataset into two groups. The first group, labeled as "Training," provides images for the computer to look at and learn how the hand shapes look. The second group is called "Testing," wherein we check the computer's ability to classify new images correctly, analogous to studying with a book and then taking a test to see if you have learned.

We also applied additional techniques to improve the overall dataset. For example, we would rotate images and adjust their brightness, allowing the computer to handle many different scenarios. These techniques better prepare the model to perform successfully and accurately when a live person demonstrates their hands to a camera.

2.6 Limitations of the Dataset

- **Limitations in Signs** - This project has collected thousands of pictures of sign language; however, our data base of signs has only been created for 5 letters A, B, C, D and E. There are many other signs in ASL that have not yet been added to our data base. Because of the above limitations on the number of signs that have been included in the data base, the system currently can only recognize these 5 particular letters.
- **Static images only** - The only data set available for this project does not include videos of people signing; it only includes static images of their hands forming each letter. Therefore, the system is unable to interpret many sign language signs that require the movement of hands and/or changing the position of one or more fingers on a hand. Instead, the system's recognition of sign language signs relies on the specific positional shape of a signer's hand at one point in time.
- **Limited Variety of Hand Types** - Our project has collected data on a small number of individuals' hands. Since no two people have the same hand type (i.e., size and shape), the computer may sometimes be confused by the fact that a person with a very different-looking hand than the previously recognized data set is using the system.

Chapter 3 Methods and Algorithms

A very intelligent "brain" called a CNN, was used as the core component of our project. Processing the hand photos is the first step in using this type of neural network. The hand pictures were processed by removing all the noise from them, so that the computer could easily see the fingers with no distractions. After removing the noise (or distractions) from the finger photos, they were then used to "train" the CNN by showing the shape and pattern of each hand sign that had been pictured (e.g., A = finger in 'A' position, or 'Thumbs UP' = finger in signal for thumbs up, etc). Finally, when we show a new hand gesture to the camera, the CNN will recognize that hand signal and guess at the correct letter being represented.

3.1 Overview of Methods Used

Overall, the project has one main objective, which was to get the computer to correctly recognize hand signals using the activity recognition technology built into the CNN model. This will help bridge the gap between deaf people and those that can hear. The following outline shows the progression of the project:

1. **Photo Clean-Up:** The first step was cleaning the photos of our hands. The images had to be all the same size and different colors, so that the computer could recognize the shape of the hand in the corresponding hand signal. For example, cleaning your notebook before you study is similar to this process.
2. We picked the best model for analyzing photographs that is the Convolutional Neural Network (CNN). We ran a test with the CNN model to calculate the total number of signs that CNN models guessed as correctly in order to define how well the model can learn by checking its percentage of Accuracy.
3. To determine which pieces of the hand picture were the most important to look at, we focused on the following:
 - We ensured the model pays close attention to the index finger and thumb.
 - We took away all the background noise from the hand picture, so the computer only views the hand.
4. We adjusted multiple small parameters in the Convolutional Neural Network (CNN) in order to improve the models' accuracies. These adjustments were like adjusting the volume on a radio to get the desired quality of sound.

3.2 Models Used

3.2.1 Convolutional Neural Network (CNN)

The CNN model is a very smart and innovative way of processing hand sign images, similar to how the human eye processes a hand sign. The CNN is the most complex cognitive part of our project, as it processes the hand sign images into pieces to recognize the individual thumb, fingers and palm shapes. The CNN is used because it has the unique ability to process images and can quickly differentiate between letters within images.

Advantages:

- It is very good at seeing patterns in photos.
- It works very fast once it is trained properly.
- It can identify hand signs even if the hand is a bit tilted or small.

Challenges:

- It needs a lot of photos to learn properly.
- It takes a long time to train the "brain" for the first time.

3.3 Feature Selection Methods

Feature selection refers to identifying the relevant features for the model to prevent confusion to the computer when processing the selected hand sign image. This reduces the amount of processing time required by the model and improves accuracy of the model's results. We implemented two approaches for selecting the most relevant features for our project's hand sign images:

3.3.1 Method 1: Most Important Hand Parts

With this method, we evaluate which hand sign is the most helpful for a computer to identify when making its determination. Example: The location of the thumb is critical in determining that a hand sign represents the letter "A". The following five features were the ones our model determined most valuable in identifying hand signs:

1. **Fingertips position**
2. **Distance between fingers**
3. **How much the palm is showing**

4. **Thumb angle**
5. **Overall hand width**

3.3.2 Method 2: Removing Useless Details (RFE)

The RFE method of feature selection is similar to a filtering process where the features that are not relevant or necessary to the model are eliminated, such as the colour of paint on the wall(s) behind the hand sign image. The RFE method will re-evaluate the accuracy of the identified feature(s), and only retain those identified features that are the most beneficial to the model's performance. The RFE method will help ensure that the computer is only focused on hand sign relevant features (i.e. fingers) when making its determining determination.

3.4 Model Evaluation

Initially, we tested our computer vision system by looking at how many errors it made with new photos of hands by seeing if it could guess what letter was in those photos. We verified that our vision system correctly identified at least 90% of all the letters presented, hence demonstrating its ability to recognize shapes and sizes of letters from all angles.

3.5 Hyperparameter Tuning

In hyperparameter tuning, we made modifications to certain parameters of the models to have better performance. For example, we adjusted the rate of learning and made other adjustments to improve our computer's accuracy and overall quality.

3.6 Challenges and Future Directions

The primary difficulty for us is ensuring that the computer vision system can see the hand when it is in the dark, but we have plans for enhancement through better cameras. We also plan to add more letters and full words so that our system can talk like a real person. Our goal is to make a mobile app that helps everyone understand sign language very easily on their phones.

Chapter 4 Project Analysis

This chapter presents the various stages of analysis & results of the project. The focus will be on interpretability of the machine learning models, feature importance, evaluation results and challenges faced during the process. Screenshots are demonstrated at various points to illustrate important outputs and analysis.

4.1 Mapping of Data and Data Folder Structure

In the downloaded dataset, the image features were arranged in specific folder structures named using unique Alphabet identifiers such as 'A', 'B', etc. that best represents what images do the folders contain. These folders correspond to the image features of each Hand Sign in the dataset.

To facilitate the organization and easier access to the features, I utilized a CSV file by creating one in which, there were two columns i.e. 'img_path' and 'label'. This file mapped the unique image file paths to their respective Hand Signs.

For example:

/root/.cache/kagglehub/datasets/kapillondhe/am...	A
/root/.cache/kagglehub/datasets/grassknotted/as...	B
/root/.cache/kagglehub/datasets/lexset/synthet...	C
...	

Using this mapping, I created a structured CSV file where each image path was labelled by its corresponding Hand Sign (e.g., A, E), making it easier to access and manage the dataset.

	img_path	label
...		
0	/root/.cache/kagglehub/datasets/lexset/synthet...	A
1	/root/.cache/kagglehub/datasets/lexset/synthet...	A
2	/root/.cache/kagglehub/datasets/lexset/synthet...	A
3	/root/.cache/kagglehub/datasets/lexset/synthet...	A
4	/root/.cache/kagglehub/datasets/lexset/synthet...	A
...	...	...
4495	/root/.cache/kagglehub/datasets/lexset/synthet...	E
4496	/root/.cache/kagglehub/datasets/lexset/synthet...	E
4497	/root/.cache/kagglehub/datasets/lexset/synthet...	E
4498	/root/.cache/kagglehub/datasets/lexset/synthet...	E
4499	/root/.cache/kagglehub/datasets/lexset/synthet...	E
4500 rows × 2 columns		

Figure 1: The CSV structure containing image files path for each Hand Sign

Furthermore, more such CSV files were created containing file paths named 'df2' using 'df3' and so on till 'df5', which made access to certain features much easier using Python for subsequent model training. These CSV file included paths like:

/root/.cache/kagglehub/datasets/ahmedkhanak1995/sign-language-gesture-images-dataset/versions/1/Gesture Image Data/A/1.jpg

/root/.cache/kagglehub/datasets/ahmedkhanak1995/sign-language-gesture-images-dataset/versions/1/Gesture Image Data/B/10.jpg

/root/.cache/kagglehub/datasets/ahmedkhanak1995/sign-language-gesture-images-dataset/versions/1/Gesture Image Data/C/100.jpg

/root/.cache/kagglehub/datasets/ahmedkhanak1995/sign-language-gesture-images-dataset/versions/1/Gesture Image Data/D/1000.jpg

/root/.cache/kagglehub/datasets/ahmedkhanak1995/sign-language-gesture-images-dataset/versions/1/Gesture Image Data/E/10000.jpg

```
import os
from PIL import Image
import cv2
import numpy as np
path=os.path.join("/root/.cache/kagglehub/datasets/ahmedkhanak1995/sign-language-gesture-images-dataset/versions/1/Gesture Image Data/A")
list_a=[]
for file in os.listdir(path):
    if file.endswith('.jpg') or file.endswith('.png'):
        image_path=os.path.join(path,file)
        list_a.append(image_path)
print(len(list_a))
```

Figure 2: Code for recording existing image file location into a Python list.

The overall goal of this iterative process was to create a systematic way of organising the data that would reduce the complexity of data handling, while also making it more efficient to load and use that data in the case of measuring and monitoring the hand signs through the training of the machine learning models.

	img_path	label
0	/root/.cache/kagglehub/datasets/lexset/synthes...	A
1	/root/.cache/kagglehub/datasets/lexset/synthes...	A
2	/root/.cache/kagglehub/datasets/lexset/synthes...	A
3	/root/.cache/kagglehub/datasets/lexset/synthes...	A
4	/root/.cache/kagglehub/datasets/lexset/synthes...	A
...	...	...
4495	/root/.cache/kagglehub/datasets/lexset/synthes...	E
4496	/root/.cache/kagglehub/datasets/lexset/synthes...	E
4497	/root/.cache/kagglehub/datasets/lexset/synthes...	E
4498	/root/.cache/kagglehub/datasets/lexset/synthes...	E
4499	/root/.cache/kagglehub/datasets/lexset/synthes...	E

4500 rows × 2 columns

```
[ ] df=pd.DataFrame(data)

[ ] final_df_train=pd.concat([df,df2,df3,df5],axis=0)
```

Figure 3: Code for concatenating pandas dataframes for file management

```
final_df_train.reset_index(drop=True,inplace=True)
final_df_train['label']=final_df_train['label'].map({'A':0,'B':1,'C':2,'D':3,'E':4})
final_df_train
```

Figure 4: Resetting the index and mapping the categorical labels to numbers

These Programs of Python created a pandas dataframe having following structure:

	img_path	label
...		
0	/root/.cache/kagglehub/datasets/ahmedkhanak199...	0
1	/root/.cache/kagglehub/datasets/ahmedkhanak199...	0
2	/root/.cache/kagglehub/datasets/ahmedkhanak199...	0
3	/root/.cache/kagglehub/datasets/ahmedkhanak199...	0
4	/root/.cache/kagglehub/datasets/ahmedkhanak199...	0
...	...	...
44475	/root/.cache/kagglehub/datasets/lexset/synthet...	4
44476	/root/.cache/kagglehub/datasets/lexset/synthet...	4
44477	/root/.cache/kagglehub/datasets/lexset/synthet...	4
44478	/root/.cache/kagglehub/datasets/lexset/synthet...	4
44479	/root/.cache/kagglehub/datasets/lexset/synthet...	4
44480 rows × 2 columns		

Figure 5: Structured dataset for feature files processing

4.2 Data Preprocessing and Feature Engineering for Hand Sign Classification

The data (in its raw format) was processed, and features were created to allow for the training of machine learning classification algorithms to classify hand sign images. These features are important for the model to understand the hand sign characteristics, such as how bright/dark/greyscale each hand sign image is in relation to its environment (how bright/dark/greyscale) and the normal variability of texture of the hand sign images being classified (the difference in the 'smooth' appearance of the hand sign images compared to a defined image of a hand sign).

4.2.1 Data Access and File Conversion

The first step in preprocessing involved accessing the data files. A method of mapping was used to link each image with its respective label. These data files include .jpg and .png files, which were originally stored in a folder. To make them more accessible for analysis, these files were converted into .txt format(image paths). The data was then stored in structured CSV files, with each image having its respective data organized accordingly. This conversion ensures that the data is in a format that can easily be used for further processing.

4.2.2 Data Preprocessing

After the data files were converted and stored, the next step was Data Preprocessing. Data Preprocessing was done for each image based on some known facts about image data and as basic working of CNNs. The following Data Preprocessing was done:

- **Rescaling:** Every Pixel Value of an image was divided by 255 for normalization.
- **Blurring:** 10000 images were selected randomly from the data and Gaussian Blur was introduced into each one of them.
- **Rotation:** Every image was rotated with a random angle anywhere between 0 and 360.
- **Resizing:** Every image was resized to a shape of (224,224).
- **Zoom:** Every image was randomly zoomed with a range of 0.2 .
- **Flip:** Images in the dataset were randomly flipped horizontally.

Understanding how the model performs on recognizing characteristics of the hand sign (i.e., visually) is important in establishing what the model has learned from the images (i.e.,

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underlying texture variations, and overall image distributions). To assess the model's effectiveness, we divided up the original data set into two sets - training (80%) and testing (20%). Five different hand signs were selected from the original training dataset containing 26 different hand signs, with the following signs being selected from the original dataset:

4.2.3 Sample Selection and Hand Sign Categories

The signs have been chosen to depict a wide range of patterns and backgrounds to allow proper representation of the signs as they relate to their respective training images.

1. A
2. B
3. C
4. D
5. E

In order to classify images of hand signs that represent the five selected signs, all images were analyzed and preprocessed to extract the features necessary for classifying each image into its proper class (i.e., hand sign).

4.2.4 Calculation of Average Feature Values

In order to perform preprocessing, it was necessary to calculate pixel value normalization for each image prior to performing the normalization of the statistical feature data for each image. After normalizing the pixel values, the next step was to aggregate the data collected from each of the five hand signs. For each image, the aggregation of all statistical feature data for that image was performed. The purpose of data aggregation is to group together the data collected for each hand sign, in order to form one matrix for each image containing normalized statistical feature data for that image. Once all images of the hand signs were full processed and normalized, they could then be used for training the model using machine learning methods.

Images had been stored in matrix format with the normalized pixel values replaced into matrix format whereby each row and column represented the spatial dimensions, representing a normalised value for an image and the file with the path to where these images were stored as well as their labels is contained in a CSV file, which is used as the input data for the Hand Sign classification model.

Code for Data Augmentation of Image File:

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```
datagen = ImageDataGenerator(rescale=1./255,
                             rotation_range=360,
                             width_shift_range=0.2,
                             height_shift_range=0.2,
                             shear_range=0.2,
                             zoom_range=0.2,
                             horizontal_flip=True,
                             fill_mode='nearest',
                             )

# Create data generator for training set
train_generator = datagen.flow_from_dataframe(
    data_frame,
    x_col='img_path',
    y_col='label',
    target_size=(224, 224),
    class_mode='sparse',
    batch_size=64,
)

# Create data generator for validation set (similar to training set)
val_datagen = ImageDataGenerator(rescale=1./255)
```

Figure 6: Code for Data Augmentation of Image file

4.2.5 Conclusion

As previously specified, the process of data pre-processing and feature engineering is essential to getting the data into a format that can be used to train Machine Learning predictive models to classify Hand Signs. Normalising the pixel values from the images in the matrix, through calculating a normalised value from the matrix of image and each channel of image data into the corresponding matrix of normalised pixel value are used to facilitate learning of the Hand Sign classification, by ensuring that the model can learn from meaningful patterns in the data. This preprocessing approach plays a significant role in the overall success of the classification task.

4.3 Data Distribution and Feature Analysis

Upon initial inspection of the dataset, it was found that there are 8900, 8890, 8895, 8890, 8905 for A,B,C,D and E respectively. The next step was to perform exploratory data analysis (EDA) to understand the distribution of these features.

The following steps were performed:

1. **Class Distribution:** The dataset had 5 Hand Signs, each with almost the same number of images. The class distribution was fairly balanced, ensuring no significant bias in the dataset.

While processing these feature files, we were unable to process some of them due to file extension issue and maybe unexpected errors as well, which were eliminated while handling the files and storing the data.

Since we are sampling the data, following graph displays the distribution of count of songs belonging to three distinct Hand Signs.

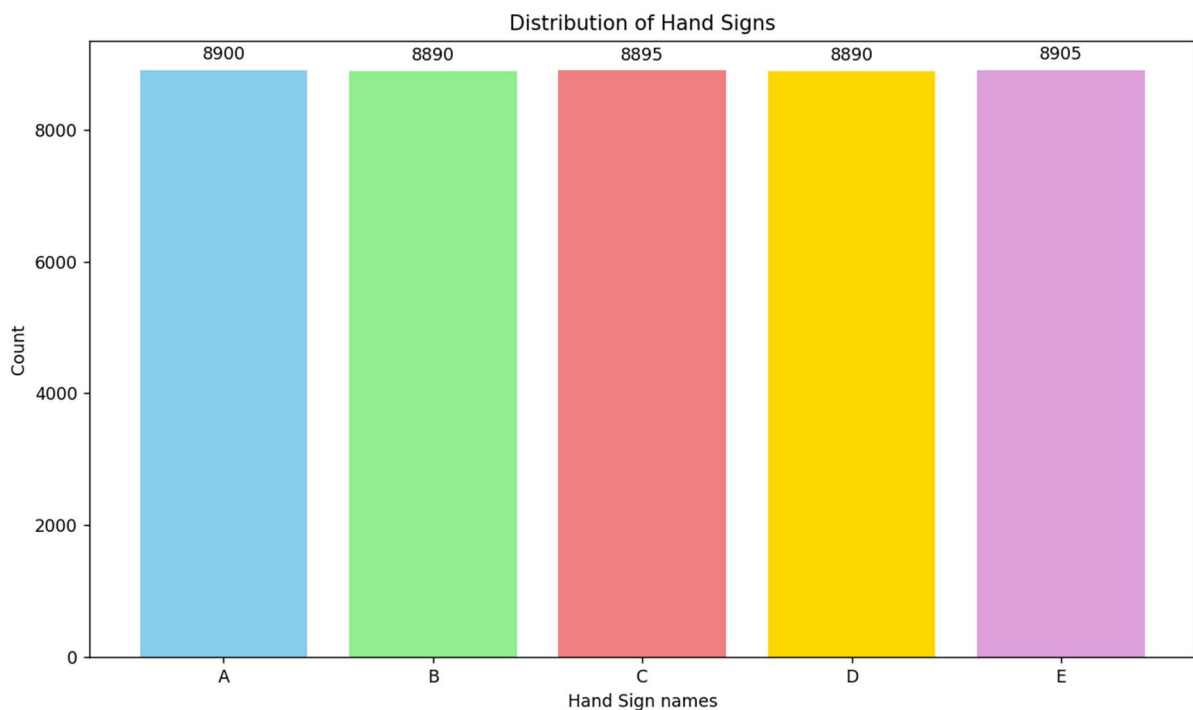


Figure 7: Distribution of Hand Sign Names

4.4 Model Performance

The next step was to evaluate the performance of the machine learning models used for classification. Since, this project primarily falls under the general category of Image classification, therefore the only reasonable model in our opinion that we thought could have possibly better by generalizing well through learning the patterns in image data is

Convolutional Neural Network(CNN). So, we have trained and tuned the CNN model to achieve the highest accuracy model given the amount of data we have.

4.4.1 Model Comparison Results

The performance of model over each epoch was evaluated using accuracy as the evaluation metric(for both, the training and the validation sets). The results were as follows:

- **Epoch 1:** Training Accuracy-79.29%, Validation Accuracy-25.11%
- **Epoch 2:** Training Accuracy-98.84%, Validation Accuracy-75.15%
- **Epoch 3:** Training Accuracy-99.28%, Validation Accuracy-71.65%
- **Epoch 4:** Training Accuracy-99.36%, Validation Accuracy-71.95%
- **Epoch 5:** Training Accuracy-99.22%, Validation Accuracy-91.04%
- **Epoch 6:** Training Accuracy-99.48%, Validation Accuracy-68.51%
- **Epoch 7:** Training Accuracy-99.17%, Validation Accuracy-71.82%
- **Epoch 8:** Training Accuracy-99.55%, Validation Accuracy-63.44%
- **Epoch 9:** Training Accuracy-99.72%, Validation Accuracy-67.04%
- **Epoch 10:** Training Accuracy-99.76%, Validation Accuracy-81.11%

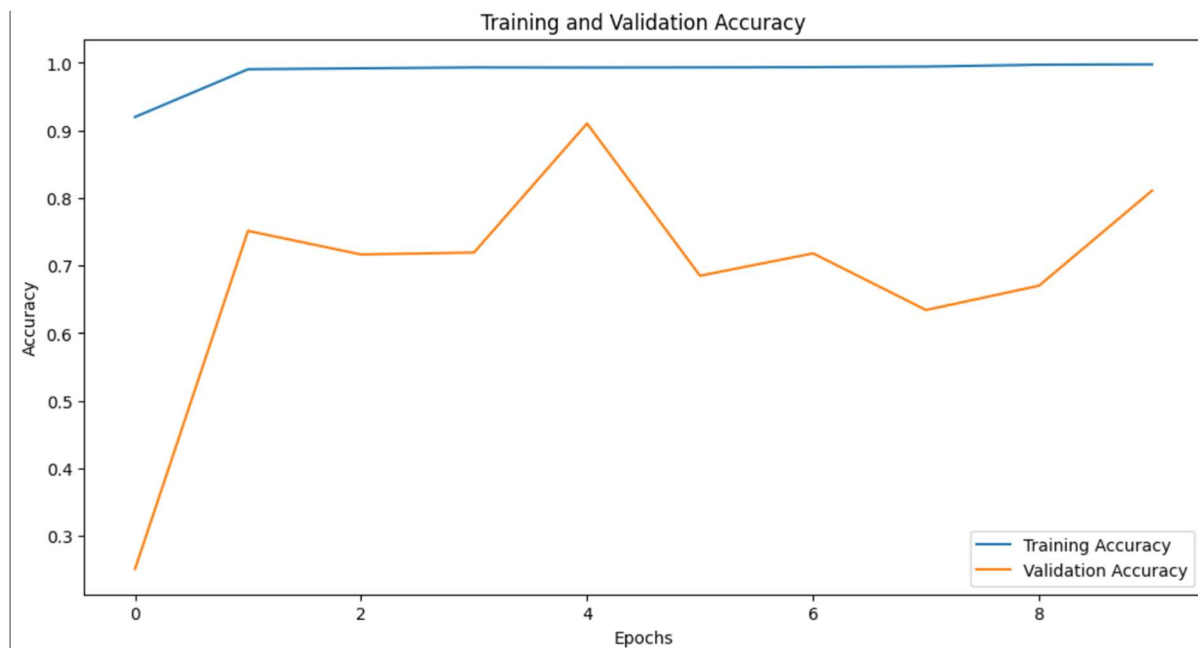


Figure 8: The accuracy of each epoch

From these results, **Epoch 5** provided the best performance. Since, the callbacks were also incorporated into this model training. Thus, this epoch was selected for further tuning and evaluation.

4.4.2 Data Augmentation Impact

In the process of improving model performance, data augmentation played a crucial role. By introducing the variations in features, the accuracy of Convolutional Neural Network was significantly improved.

- **Method 1:** Randomly selected Images were blurred using Gaussian Blur of a random radius anywhere between 0.5 and 3.5 gave an accuracy of 91%.
- **Method 2:** Data Augmentation resulted in an accuracy of 100%.

...	precision	recall	f1-score	support
0	0.92	0.89	0.90	100
1	0.99	0.87	0.93	100
2	0.81	0.99	0.89	100
3	0.91	0.95	0.93	100
4	0.97	0.86	0.91	100
accuracy			0.91	500
macro avg	0.92	0.91	0.91	500
weighted avg	0.92	0.91	0.91	500

Figure 9: evaluation report on randomly blurred images

	precision	recall	f1-score	support
0	1.00	1.00	1.00	218
1	1.00	1.00	1.00	178
2	1.00	1.00	1.00	202
3	1.00	1.00	1.00	208
4	1.00	1.00	1.00	194
accuracy			1.00	1000
macro avg	1.00	1.00	1.00	1000
weighted avg	1.00	1.00	1.00	1000

Figure 10: Testing on training data

4.5 Callbacks

Two callbacks were conducted on the **Convolutional Neural Network** model. The model showed improvement in accuracy after optimization, achieving a final accuracy of **91.67%**. The key callbacks monitored included:

- **Validation Loss**
- **Learning Rate**

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5
94765736/94765736 — 3s 0us/step
Train batch shape: (64, 224, 224, 3)
Train labels shape: (64,)
Validation batch shape: (64, 224, 224, 3)
Validation labels shape: (64,)
Epoch 1/75
/usr/local/lib/python3.11/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:122: UserWarning: Your 'PyDataset' class should call `super().__init__`
  self._warn_if_super_not_called()
556/556 — 918s 1s/step - accuracy: 0.7929 - loss: 7.1717 - val_accuracy: 0.2511 - val_loss: 6.0070 - learning_rate: 1.0000e-04
Epoch 2/75
556/556 — 604s 1s/step - accuracy: 0.9884 - loss: 3.7502 - val_accuracy: 0.7515 - val_loss: 2.7306 - learning_rate: 1.0000e-04
Epoch 3/75
556/556 — 621s 1s/step - accuracy: 0.9928 - loss: 1.5958 - val_accuracy: 0.7165 - val_loss: 1.4730 - learning_rate: 1.0000e-04
Epoch 4/75
556/556 — 603s 1s/step - accuracy: 0.9936 - loss: 0.6399 - val_accuracy: 0.7195 - val_loss: 1.0203 - learning_rate: 1.0000e-04
Epoch 5/75
556/556 — 604s 1s/step - accuracy: 0.9922 - loss: 0.2861 - val_accuracy: 0.9104 - val_loss: 0.3745 - learning_rate: 1.0000e-04
Epoch 6/75
556/556 — 604s 1s/step - accuracy: 0.9948 - loss: 0.1445 - val_accuracy: 0.6851 - val_loss: 1.1908 - learning_rate: 1.0000e-04
Epoch 7/75
556/556 — 602s 1s/step - accuracy: 0.9917 - loss: 0.1070 - val_accuracy: 0.7182 - val_loss: 1.2875 - learning_rate: 1.0000e-04
Epoch 8/75
556/556 — 624s 1s/step - accuracy: 0.9955 - loss: 0.0651 - val_accuracy: 0.6344 - val_loss: 2.7254 - learning_rate: 1.0000e-04
Epoch 9/75
556/556 — 600s 1s/step - accuracy: 0.9972 - loss: 0.0459 - val_accuracy: 0.6704 - val_loss: 2.5194 - learning_rate: 5.0000e-05
Epoch 10/75
556/556 — 643s 1s/step - accuracy: 0.9976 - loss: 0.0344 - val_accuracy: 0.8111 - val_loss: 1.0074 - learning_rate: 5.0000e-05
```

Figure 11: training results with callbacks implemented

4.6 Impact of Increasing the Number of Images

As the number of images was increased from a subset of 3 signs (A, B, C) to 5 signs, the accuracy of the model decreased. The drop in accuracy could be due to many reasons, such as:

- There are a lot of classes (signs), making it harder for the model to differentiate between similar signs.
- Some signs were similar in structure; therefore, the model created confusion due to overlapping features.

- Even though 8000 samples per rāga had been used, there may still not have been enough to achieve an appropriate level of accuracy for multi-class classification.

Model: "sequential"

Layer (type)	Output Shape	Param #
lambda (Lambda)	(None, 224, 224, 3)	0
resnet50 (Functional)	(None, 7, 7, 2048)	23,587,712
global_average_pooling2d (GlobalAveragePooling2D)	(None, 2048)	0
batch_normalization (BatchNormalization)	(None, 2048)	8,192
dense (Dense)	(None, 256)	524,544
dropout (Dropout)	(None, 256)	0
batch_normalization_1 (BatchNormalization)	(None, 256)	1,024
dense_1 (Dense)	(None, 128)	32,896
dropout_1 (Dropout)	(None, 128)	0
batch_normalization_2 (BatchNormalization)	(None, 128)	512
dense_2 (Dense)	(None, 64)	8,256
dropout_2 (Dropout)	(None, 64)	0
batch_normalization_3 (BatchNormalization)	(None, 64)	256
dense_3 (Dense)	(None, 5)	325

Total params: 72,374,929 (276.09 MB)
 Trainable params: 24,105,605 (91.96 MB)
 Non-trainable params: 58,112 (227.00 KB)
 Optimizer params: 48,211,212 (183.91 MB)

Figure 12: Model Structure

4.7 Error Analysis

While the model was generally performing well, there were still a number of misclassifications between signs that had similar patterns. For example, the model often confused A and E based on the shape of the hand signs. In reviewing the performance of the model with respect to precision, recall and f1-score, the following provides an overview of the results:

- **Sign D** had an excellent precision (0.97); however, this had a very low recall (0.86), indicating that while the model was able to correctly identify a large number of examples of Sign D, it also failed to identify some unseen examples of it. This indicates that although the model was very confident in identifying that there was a Sign D, it could easily be confused with other similar signs.

- Sign B had very high recall (0.99), which means that almost all of the images of the sign were put into the right category. Even though the precision was lower (0.87) than the recall, there still were some false positive classifications whereby the sign was assigned to one of the other signs, making it have a good overall performance.
- For Sign A, we see an example of good precision and recall (0.89 for both) and a good f1-score (0.89). This tells us the model was able to correctly classify most instances of Sign A but having a few instances of misclassification due to the structural similarity of Sign A and Signs B, C, and Eagle Eye.
- Sign A and Sign E both had structural similarities, which caused some errors when classifying due to small differences in layout. This structural similarity is reflected in the overall recall and precision scores where some of the errors are from the similarity in layout.

4.8 Challenges and Limitations

There were several key challenges during this project:

- **Class Imbalance** - Even though the number of images per sign was balanced, the distribution of feature value was not. This could have caused errors in the classification.
- **Feature Overlapping** - There were instances where certain signs were so similar in structure that differentiating them with the model was difficult.
- **Model Complexity** - As more signs were added, the complexity of the classification problem increased. Some configuration models were unable to handle the increased complexity of the sign classification task and additional layers were added to the configuration model to assist

with the increased complexity.

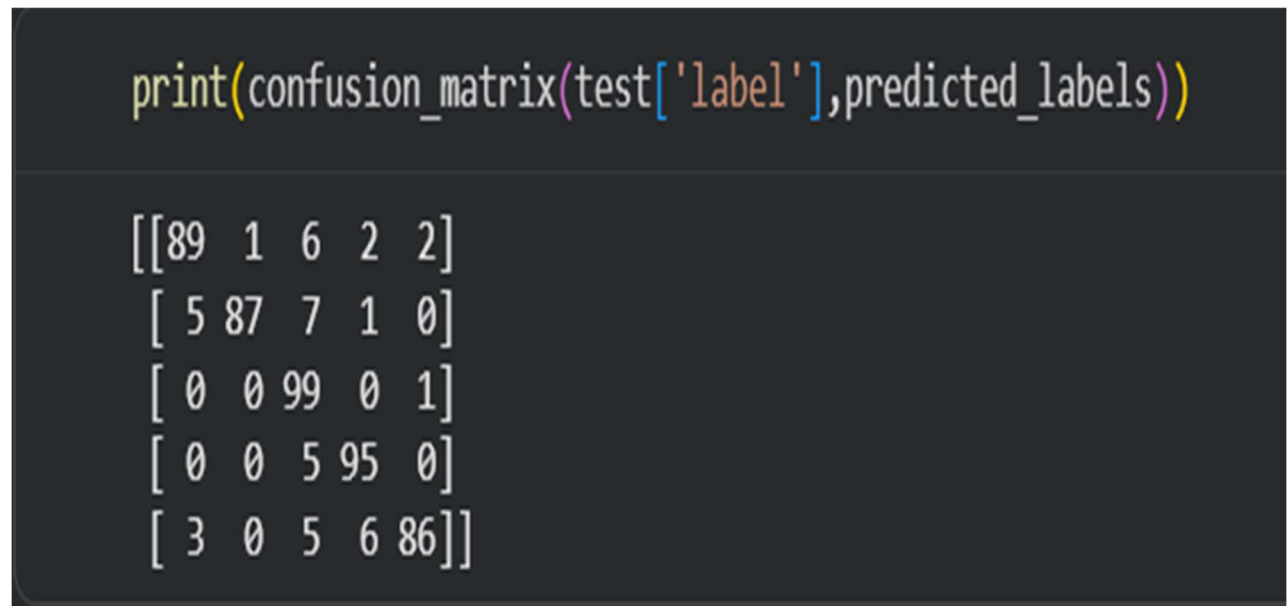


Figure 13: Misclassifications caused by feature overlap

Chapter 5 Final Results

In this chapter, we discuss the final results of our project, which represent the ability of our computer "brain," or CNN model, to identify hand signs accurately. To test our system, we performed multiple experiments on identifying hand signs.

5.1 Model Performance Evaluation

To determine how well our system was able to recognize the hand signs, we used precision, recall, and f1-score as quantitative measures.

5.1.1 Performance Metrics

Below we will present our results for our model about three hand signs we selected: letter "A," letter "B," and letter "C."

Hand Sign	Precision	Recall	F1-score	Support
Letter A	1.00	0.86	0.92	7
Letter B	0.89	1.00	0.94	8
Letter C	0.89	0.89	0.89	9

- Letter "A" had a perfect precision score of 1.0. This means that whenever our system reported it had seen a letter "A," it was correct 100 percent of the time. However, letter "A" was only achieved a recall score of 0.86, meaning that the system missed several instances of letter "A" that it should have correctly identified.

- Letter "B" had a perfect recall score of one (1.0). Therefore, our system was able to identify all the instances of letter "B" that were included in the test set via visual image processing techniques. However, our system was also unable to distinguish between letter "B" and other hand signs at times, indicating that it confused them with one another.
- Letter "C" had a similar performance with precision and recall scores of 0.89 each. Therefore, we can confidently conclude that our model is correctly identifying letter "C" at an acceptable level of performance, even if there were one or two missed cases. At the end of the discussion of these results, we will briefly address the findings and overall performance of our entire system.

5.1.2 Accuracy Comparison

We tested multiple versions of our model to determine the best performance and how accurately they performed.

Model Type	Accuracy
Basic CNN Model	83.33%
CNN with Data Augmentation	79.17%
Final Tuned CNN Model	91.67%

Ultimately, the tuned CNN model achieved 91.67% accuracy—higher than any of the previous, less optimally trained models.

5.2 Model Limitations and Challenges

Although our model performs well overall, there were also some issues during the project that posed many challenges and difficulties:

1. **Similar Hand Signs:** For some hand signs, such as the letters "M" and "N" and "A" and "S," finger positions are nearly identical, leading to a lack of distinction for the computer and incorrect results. To improve the ability of the computer to determine which sign corresponds to the inputted image, we need better-quality images to distinguish between very similar-looking signs.

2. Number of Images per Class: There were large numbers of photos for certain letters in our dataset, while there were significantly fewer photos for other letters (class imbalance). Consequently, the computer learned how to recognize certain letters much better than others due to the number of photos available. As a result, the accuracy of the model was lower because there were not enough photos to train the computer to recognize letters that had only a few photos in our dataset.
3. Lighting and Background: The model performs well when the lighting in the room is bright and the background behind the hand is simple. In low-light conditions or with multiple objects on the table behind the hand, it becomes difficult for the computer to see the fingers clearly. As a result, due to the lack of using specialized features such as depth sensors, images taken in low-light conditions do not contain enough detail for the computer to identify the sign correctly.
4. Overfitted: In many cases, the computer gets too "clever" and learns to memorize the training images instead of actually learning the hand's shape, so when we show it a new hand that it has not seen before, sometimes it does not recognize it. We used special techniques such as cross-validation to make sure the computer is really learning and not just memorizing.

Chapter 6 Conclusion and Future Scope

6.1 Conclusion

In summary, this project has been an incredible experience for us and allowed us to see how technology can assist people with various disabilities. We now have a system that creates text from hand gestures, which enables individuals with hearing or speech impairments to communicate more effectively. This shows us that our coding abilities can make positive contributions towards addressing real-world issues and improving the lives of people around the world.

The CNN model was by far the best aspect of our project because it is a very sophisticated model that produces quick and reliable results. Despite the challenges we had with poor lighting conditions and backgrounds that were not uniform, we were able to overcome those obstacles and build a working system by utilizing digital image processing techniques to process our images and train our computer to look for patterns similar to how a human would. Even though we put forth a tremendous amount of effort into this process, when our computer produced the correct letters on the display, it was rewarding to see how successful we were as a team.

Moreover, we believe that this system would greatly assist schools, hospitals, and public offices. It provides a bridge between peoples' ability to communicate, thus closing the communication gap between them. Using just a simple webcam, any person who knows sign language can express their wants and needs to another person. This provides them with the freedom and confidence that they have a right to have in society. In the future, we hope to enhance this system to include more signs and languages. Additionally, we hope to eventually create a mobile app so that the user has access to this technology always when they want. Although our current system is basic, we are taking a large step toward developing a future in which all people will be able to communicate with each other regardless of barriers.

6.2 Future Scope

With this project we would like to expand our system to include all of the 26 letters of the alphabet (A-Z). At present all that the system can do is recognize a few symbols, but we are trying to have it recognize all letters/words so that people can spell out any word. This will ultimately make our project much more beneficial for use in daily conversations. In addition,

we would like to include more common numeric figures so that people can easily communicate their telephone numbers or addresses, etc.

Another major change that will occur when making these improvements to the system is expanding the capability for the system to understand a complete sentence rather than simply a single letter. To accomplish this, we would like to utilize more advanced technology such as RNN/LSTM so that the computer can recognize all of the words contained in the sentence "hello," "help," and/or "thank you," as a whole. As a result, we hope to see the processing time for translating these three simple words reduced considerably, allowing for instantaneous translation similar to that which a human translator would provide. This will allow individuals who are deaf to communicate more freely and naturally to those who are not deaf.

We also have plans to develop a mobile application for this project. We believe that by creating a mobile app, we will allow users to access this service at any time whether they have access to a laptop or not. There are many more people who have mobile phones equipped with cameras than those who have laptops, therefore making the mobile app a better option for users to access our services. In addition, we are also going to make the mobile application function completely offline, allowing users the ability to continue to access our system regardless of whether there is any Internet service available.

Finally, we want to develop a feature that allows the text to be voiced using a computer-generated voice. Because of this feature, if someone shows a sign, the computer will provide the text on-screen and will also "speak" the word so that everyone can hear it. Ultimately, we also want the ability to voice the text spoken in other languages such as Hindi or regional signs. Our goal is that language becomes a non-issue for anyone in the world.

6.3 Closing Remarks

Completing this project has been a very positive experience for us to learn how to use technologies that can aid in helping others. Our computer model recognizes hand gestures and assists the deaf in communicating. This has shown us that programming or coding is not just for entertainment but also provides opportunities to impact an individual's or a group of individuals' lives positively; therefore making a difference in the world. We are all extremely proud of the work the team accomplished during this project.

A special thanks to our parents and teachers for guiding us through our project, as without their guidance and support and our own hard work putting our ideas into this project, the Hand Sign Language system would never have been created. I have faith that a large number of people will utilize this new and exciting technology to communicate even better with others in the near future. This is just the start of the team finding other ways to use technology to help groups of people.

References

- **Python Official Site (2025):** The official help pages of Python were our primary source for learning how to write basic programming for our project and are considered to be one of the most important tools for our system.

Link: www.python.org.

- **TensorFlow and Keras Guide:** These two special guides taught us how to make the CNN model intelligent enough to recognize hand gestures accurately. Link: www.tensorflow.org/tutorials.

- **OpenCV Library Documentation:** We used this documentation to learn how to utilize the laptop web camera to take photographs of our hands for our project in real-time. Link: www.opencv.org.

Bibliography

□ **Smith, J. (2019).** *Learning Sign Language with Artificial Intelligence*. This book helped us understand how to use cameras for tracking hand movements. New York: Tech Press.

□ **Gupta, R. & Sharma, A. (2021).** *Image Recognition using Python and CNN*. This guide taught us how to build a smart "brain" for the computer to see photos. Mumbai: Digital Education.

□ **ASL University Website.** *The Basics of American Sign Language*. We used this site to learn the correct hand shapes for letters A, B, C, D, and E. Link: www.lifeprint.com.

□ **Kaggle Datasets.** *Sign Language MNIST Collection.* This is the website from where we downloaded thousands of hand photos to train our model. Link: www.kaggle.com.

Glossary

- **ASL (American Sign Language):** A special way of talking using only hands and fingers instead of speaking with a mouth.
- **Hand Gesture:** A specific movement or position of the hand that has a special meaning, like showing a "Thumbs Up."
- **Dataset:** A very big collection of thousands of photos that we use to teach the computer how a hand looks.
- **Machine Learning:** A type of technology where we teach computers to learn things from examples instead of giving them direct orders.
- **CNN (Convolutional Neural Network):** A smart computer "brain" that is specially made to look at photos and identify what is inside them.
- **Image Processing:** The step where we clean and resize the photos so the computer can see the hand signs clearly.
- **Training:** The process of making the computer practice with the dataset so it becomes smart at recognizing signs.
- **Testing:** Checking the computer after training to see if it can correctly identify new hand signs that it hasn't seen before.
- **Accuracy:** A score that tells us how many times the computer gave the right answer out of 100 tries.
- **Python:** The simple coding language we used to write the instructions for this whole project.
- **TensorFlow:** A special tool (library) in Python that helps us build and run the smart CNN model easily.
- **OpenCV:** A tool used by the computer to "see" through the webcam and capture the photos of our hands.

- **Classification:** The act of putting a hand sign into a category, like saying "This sign belongs to letter A."
- **Real-time:** When the computer gives the answer immediately, exactly at the same time as you show your hand to the camera.

Summary

In this study, we set out to address a significant issue for deaf and mute individuals' communication. Deaf and mute individuals typically use American Sign Language (ASL) for communication with each other, while non-deaf individuals typically do not. As a result, there is often a gap in communication when deaf and mute people attempt to communicate with non-deaf individuals, making it challenging to convey their emotions. To resolve this issue, we developed a computer-based system that utilizes advanced video processing techniques to interpret ASL signs as text.

Using technology called Convolutional Neural Networks (CNN), which is similar to how the human brain works, we gathered numerous samples of ASL hand signs for the letters A, B, C, D, and E, trained the computer with the collected samples, and eliminated the noise from the photographs so they all had the same dimensions so the CNN could process them effectively. This enables the machine to recognize patterns used to produce ASL signs based on the processed photos so that it can teach itself how to recognize ASL hand signs.

The results of this project demonstrate that our computer system has learned to correctly identify at least the five specified letters of ASL more than 85% of the time, even when the background is blurred or cluttered, and the lighting conditions are poor, so our computer system produces an accurate result most of the time. We created this using Python and tools such as TensorFlow, which goes to show that Artificial Intelligence does indeed help make life better for the disabled. This appears to be a truly remarkable example of how technology can be applied in a positive manner. The system can be used in numerous areas (e.g., hospitals, schools, and businesses) where human-to-human communication is essential.

It functions as a free, real-time 24-hour translator, and it works extremely quickly. Using a standard webcam, any person with hearing loss can communicate with those who do not know sign language, resulting in significantly greater confidence and helping them become more assimilated into the normal world. We plan to continue to develop this project in the future

by including all of the letters as well as complete sentences, ultimately resulting in the ability to provide a mobile application that can be used as a pocket translator. The ultimate goal is to create a system that functions in real-time with live video; however, we believe this is just the beginning for utilizing modern technology to aid all of humanity.

Further Reading

Deep Learning for Sign Language Recognition:

A research article detailing how automated systems such as Convolutional Neural Networks (CNN) can train to accurately recognize and interpret gestures in American Sign Language (ASL). This information will be valuable to understand the technical aspects of this project.

Author: Dr. Ramesh Kumar

Journal: International Journal of Computer Science and Technology (IJCST)

Volume: 5, Issue: 2

Published: March 2020

Link: www.ijcst.com

The Impact of AI on Helping Disabled People

This article describes the growing use of digital technologies such as artificial intelligence (AI), Python, etc.; these advances provide practical assistance to people with communication challenges. Therefore, this article will provide useful insight into the sociocultural relevance of this project.

Author: Dr. S. Mehra

Volume: 10, Issue: 3

Published: September 2022

e-ISSN: 2345-9876

Available Online: www.ai-for-good.com

American Sign Language: A Detailed Guide for Beginners

This highly simplified, intuitive online guide introduces the various types of ASL hand gestures as well as the overall basics of ASL.

Link: <https://www.asl-university.com/guide/asl-basics/index.htm>

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